

Question Number	Answer	Additional guidance	Mark
2(a)	• (adding a stain) allows structures to be seen / {differentiate / increase contrast} between structures	e.g. chromosomes, DNA, lignin, organelles etc ACCEPT show detail more clearly ACCEPT converse argument for specimens which are not stained ignore to make specimens more visible	(1)

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2(b)	An explanation that includes four of the following points: • cellulose molecules can be organised into microfibrils. (1) • (due to layers of) {cellulose / microfibrils} arranged in {different directions / a mesh} (1) • {hydrogen bonds in cellulose molecules / hydrogen bonds in microfibrils / layers (of microfibrils) / lignin / secondary thickening / pectate} add {strength / support / stability} (1) • {(cellulose) microfibrils / mesh} held together by {pectin / pectate} (1) • lignin makes {outside of xylem vessels / cell wall} impermeable (to water) (1) • the structure of lignin described (1)	ACCEPT lignin arrangement contributes to {rigidity / flexibility} (of vessel) ACCEPT cellulose microfibrils can't slide over each other ACCEPT lignin makes {cell walls / xylem} waterproof e.g. rings, spirals, ladders	(4)